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INFORMATION SECURITY

PROJECT DOCUMENTATION

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1. Introduction

In modern networks, secure communication is essential. However, attackers can exploit vulnerabilities and intercept communication. One such attack is the Man-in-the-Middle (MITM) attack, where an attacker secretly intercepts and modifies communication between two parties.

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2. Problem Statement

In unsecured communication systems, data can be intercepted and altered by attackers without the knowledge of users. This compromises confidentiality and integrity of information.

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3. Objectives

- To simulate MITM attack using C++
- To demonstrate message interception and modification
- To implement encryption as a prevention method

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4. Methodology

Step 1:

Create client, server, and attacker programs

Step 2:

Establish TCP socket communication

Step 3:

Intercept messages using attacker

Step 4:

Modify and forward messages

Step 5:

Apply encryption for prevention

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5. System Design

Components:

- Client → sends message
- Attacker → intercepts
- Server → receives

Flow:

Client → Attacker → Server

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6. Implementation

- Language: C++
- Library: Winsock
- Protocol: TCP

Programs:

- client.cpp
- server.cpp
- attacker.cpp

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7. Results

Without Encryption:

- Attacker reads message
- Attacker modifies message

With Encryption:

- Attacker sees unreadable data
- Server successfully decrypts

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8. Prevention Techniques

- Encryption (XOR in project)
- SSL/TLS
- Digital certificates
- Authentication mechanisms

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9. Limitations

- Runs only on local system
- Uses simple encryption (not strong)
- No real-world network complexity

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10. Conclusion

The project successfully demonstrates how MITM attacks work and how encryption can prevent data interception. It highlights the importance of secure communication in modern systems.

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11. Future Work

- Implement stronger encryption (AES)
- Use real network environment
- Add graphical interface